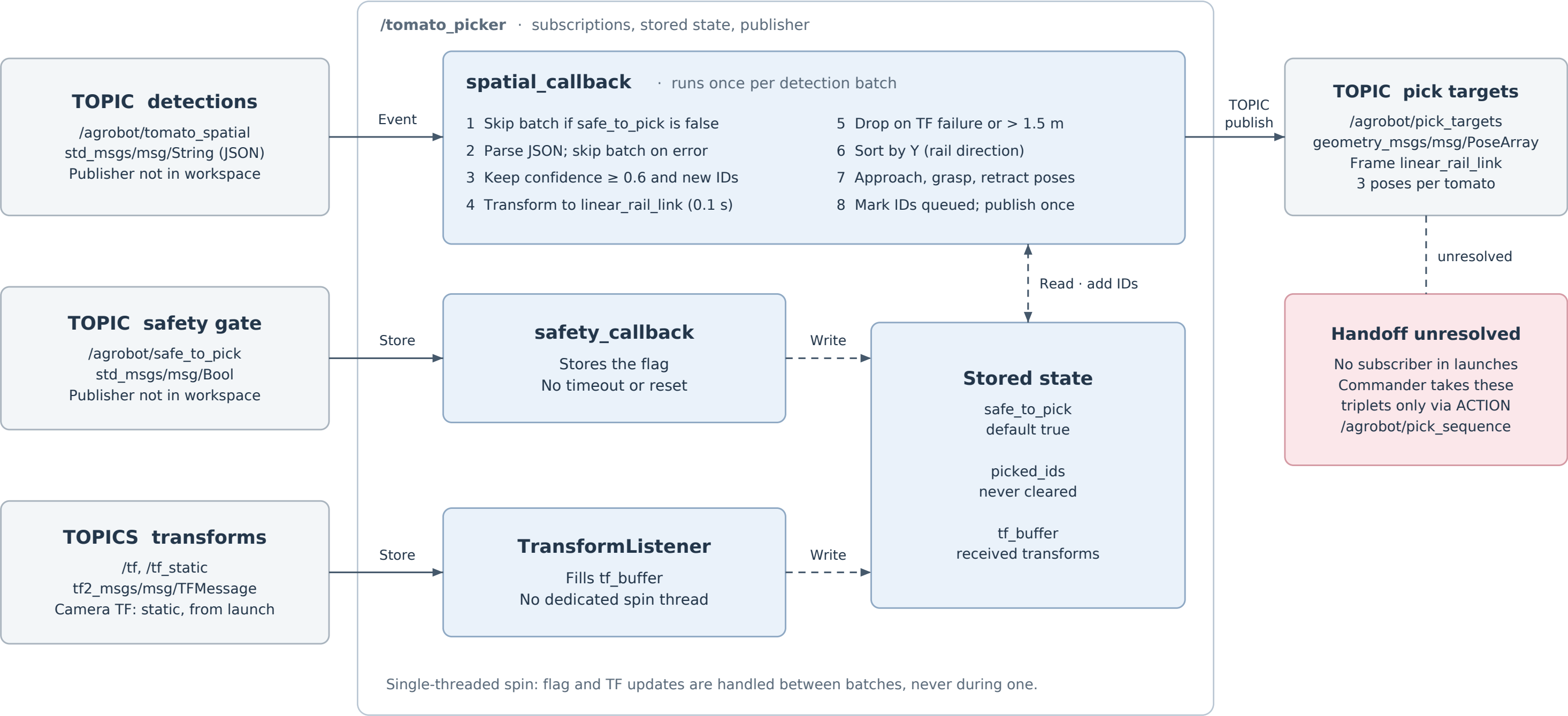


Level 2 — /tomato_picker detection to pick targets

robot_commander / tomato_picker.py · ROS node /tomato_picker · launched by commander.launch.py (no parameters)
Source-derived · rclpy.spin, single-threaded · TF listener without its own spin thread

Blue: node internals · Gray: ROS interfaces · Rose: unresolved handoff
Solid: ROS communication · Dashed: shared state or unresolved link



IDs are marked queued when published; no pick result returns to this node. No bin pose is added.
Approach = radius + 5 cm and retract = radius + 15 cm back along the base-to-tomato vector; gripper +Z is aligned to that vector.

Source evidence

Agrobot/src/robot_commander/src/tomato_picker.py:58-87 — TF listener, subscriptions, publisher; 254-258 — rclpy.spin; 51-53, 205 — thresholds, standoffs, reach.
Same file 92-95 — flag stored; 100-174 — gate, JSON parse, filter, transform, reach, sort, poses, picked_ids, publish.
Same file 179-200 — transform to linear_rail_link with 0.1 s timeout; 214-249 — approach vector, waypoints, orientation.
Agrobot/src/robot_commander/launch/commander.launch.py:14-36 — picker and camera static TF; /opt/ros/jazzy/.../tf2_ros/transform_listener.py:56 — spin_thread=False default.